Extracting Domain Specific Parts from General Purpose Programs

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Optimizing the matmul: One kernel to rule them all!

```
FOR I = 1, N:
  FOR J = 1, N:
    FOR K = 1, N:
      C[I, J] += A[I, K] * B[K, J]
    ENDFOR
  ENDFOR
ENDFOR
```

Optimizing the matmul: One kernel to rule them all!

```
FOR I = 1, N:                  ----- Dim 0
  FOR J = 1, N:                ----- Dim 1
    FOR K = 1, N:              ----- Dim 2
      C[I, J] += A[I, K] * B[K, J]
    ENDFOR
  ENDFOR
ENDFOR
```

}

Different tiling combos possible
eg. tile (0, 1) OR tile (0) OR tile (1, 0)

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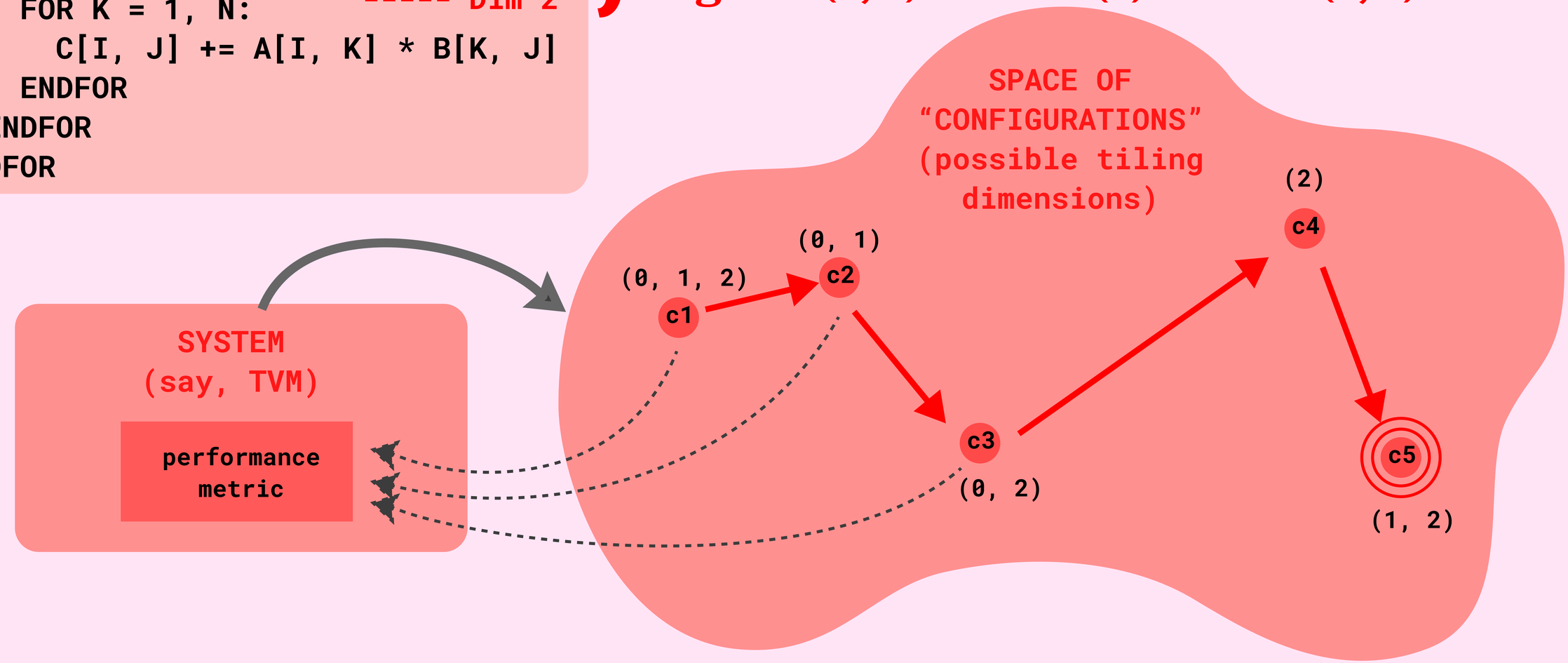

And that’s not it!

Domain specific languages try to specialize for heavily-used operations in that domain so that they’re fast. Some optimizations include

- Reductions and scans
- Layout optimization
- Graph-level rewrites

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- **Reductions and scans**
- **Layout optimization**
- **Graph-level rewrites**

However, they are hard to get used to and write code in, and are quite rigid.

```
// Are you ready? We're going to use all of the features abo
Func gradient_fast("gradient_fast");
gradient_fast(x, y) = x + y;

// We'll process 64x64 tiles in parallel.
Var x_outer, y_outer, x_inner, y_inner, tile_index;
gradient_fast
    .tile(x, y, x_outer, y_outer, x_inner, y_inner, 64, 64)
    .fuse(x_outer, y_outer, tile_index)
    .parallel(tile_index);
```

```
@tvm.script.ir_module
class MyModule:
    @T.prim_func
    def mm_relu(A: T.Buffer((128, 128), "float32"),
                B: T.Buffer((128, 128), "float32"),
                C: T.Buffer((128, 128), "float32")):
        Y = T.alloc_buffer((128, 128), dtype="float32")
        for i, j, k in T.grid(128, 128, 128):
            with T.sblock("Y"):
                vi = T.axis.spatial(128, i)
                vj = T.axis.spatial(128, j)
                vk = T.axis.reduce(128, k)
                with T.init():
                    Y[vi, vj] = T.float32(0)
                Y[vi, vj] = Y[vi, vj] + A[vi, vk] * B[vk, vj]
    for i, j in T.grid(128, 128):
        with T.sblock("C"):
            vi = T.axis.spatial(128, i)
            vj = T.axis.spatial(128, j)
            C[vi, vj] = T.max(Y[vi, vj], T.float32(0))
```

Our proposed solution

Extract domain specific parts from general purpose programs

Then use the domain specific compiler to optimize those parts!

Why is this a hard problem?
Why not replace these with hand-tuned libs?

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FOR I = 1, N:
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ENDFOR
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Why not replace these with hand-tuned libs?

```
FOR I = 1, N:
  Arr[I] = Arr[I]+1
  FOR J = i, N:
    FOR K = 1, N:
      // other code
      C[I, J] += A[I, K] * B[K, J]
      T[i] = C[I, J]
    ENDFOR
    // more code
  ENDFOR
ENDFOR
```

The major challenge is extracting hidden (latent) representation, which programmers can't do manually

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